

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education
Advanced Subsidiary Level and Advanced Level

PHYSICS

9702/01

Paper 1 Multiple Choice

October/November 2005

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions.

For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

This document consists of **19** printed pages and **1** blank page.



Data

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space,

$$\varepsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall,

$$g = 9.81 \text{ m s}^{-2}$$

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

simple harmonic motion,

$$a = -\omega^2 x$$

velocity of particle in s.h.m.,

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2}QV$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

critical density matter of the Universe,

$$\rho_0 = \frac{3H_0^2}{8\pi G}$$

equation of continuity,

$$Av = \text{constant}$$

Bernoulli equation (simplified),

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Stokes' law,

$$F = Ar\eta v$$

Reynolds' number,

$$R_e = \frac{\rho v r}{\eta}$$

drag force in turbulent flow,

$$F = Br^2 \rho v^2$$

1 Which pair of units are both SI base units?

- A ampere, degree celsius
- B ampere, kelvin
- C coulomb, degree celsius
- D coulomb, kelvin

2 The prefix 'centi' indicates $\times 10^{-2}$.

Which line in the table correctly indicates the prefixes micro, nano and pico?

	$\times 10^{-12}$	$\times 10^{-9}$	$\times 10^{-6}$
A	nano	micro	pico
B	nano	pico	micro
C	pico	nano	micro
D	pico	micro	nano

3 Which expression involving base units is equivalent to the volt?

- A $\text{kg m}^2 \text{s}^{-1} \text{A}^{-1}$
- B $\text{kg m s}^{-2} \text{A}$
- C $\text{kg m}^2 \text{s}^{-1} \text{A}$
- D $\text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$

4 A steel rule can be read to the nearest millimetre. It is used to measure the length of a bar whose true length is 895 mm. Repeated measurements give the following readings.

length / mm 892, 891, 892, 891, 891, 892

Are the readings accurate and precise to within 1 mm?

	results are accurate to within 1 mm	results are precise to within 1 mm
A	no	no
B	no	yes
C	yes	no
D	yes	yes

- 5 The density of the material of a rectangular block is determined by measuring the mass and linear dimensions of the block. The table shows the results obtained, together with their uncertainties.

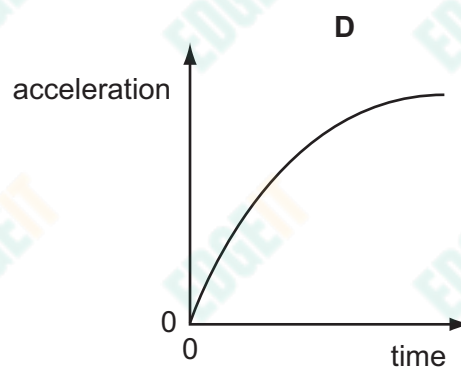
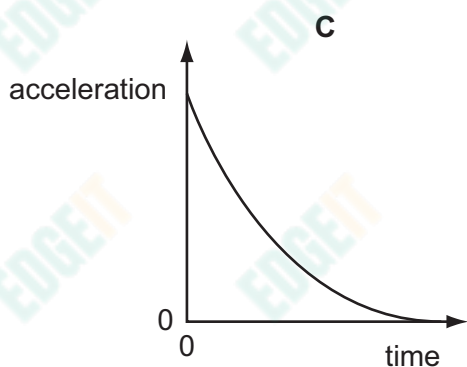
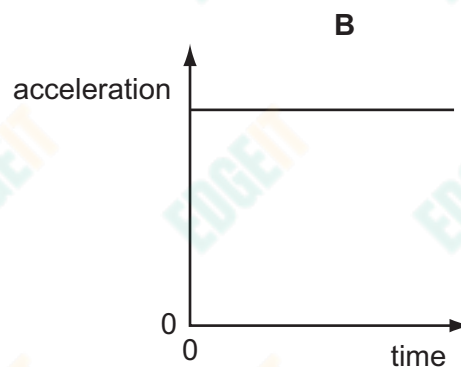
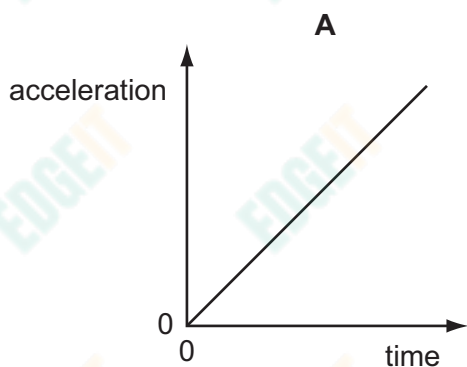
mass	=	$(25.0 \pm 0.1)\text{g}$
length	=	$(5.00 \pm 0.01)\text{cm}$
breadth	=	$(2.00 \pm 0.01)\text{cm}$
height	=	$(1.00 \pm 0.01)\text{cm}$

The density is calculated to be 2.50 g cm^{-3} .

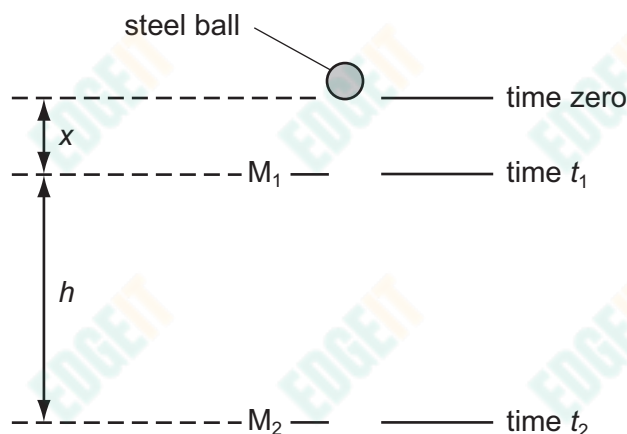
What is the uncertainty in this result?

- A $\pm 0.01\text{ g cm}^{-3}$ B $\pm 0.02\text{ g cm}^{-3}$ C $\pm 0.05\text{ g cm}^{-3}$ D $\pm 0.13\text{ g cm}^{-3}$
- 6 A football is dropped from the top of a tall building.

Which acceleration-time graph best represents the motion of the football through the air?



- 7 Two markers M_1 and M_2 are set up a vertical distance h apart.

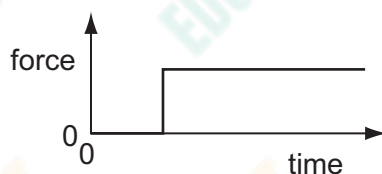


A steel ball is released at time zero from a point a distance x above M_1 . The ball reaches M_1 at time t_1 and reaches M_2 at time t_2 . The acceleration of the ball is constant.

Which expression gives the acceleration of the ball?

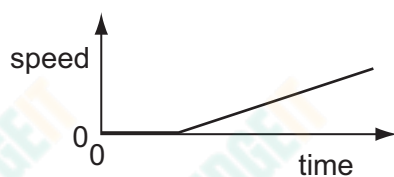
- A $\frac{2h}{t_2^2}$ B $\frac{2h}{(t_2 + t_1)}$ C $\frac{2h}{(t_2 - t_1)^2}$ D $\frac{2h}{(t_2^2 - t_1^2)}$

- 8 A car driver sharply presses down the accelerator when the traffic lights go green. The resultant horizontal force acting on the car varies with time as shown.



Which graph shows the variation with time of the speed of the car?

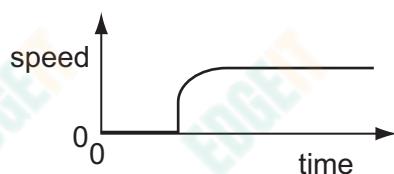
A



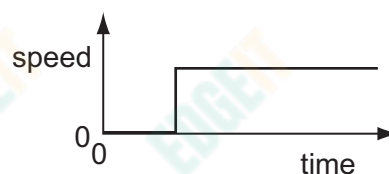
B



C



D



9 Which is a statement of the principle of conservation of momentum?

- A A force is equal to the rate of change of momentum of the body upon which it acts.
- B In a perfectly elastic collision, the relative momentum of the bodies before impact is equal to their relative momentum after impact.
- C The momentum of a body is the product of the mass of the body and its velocity.
- D The total momentum of a system of interacting bodies remains constant, providing no external force acts.

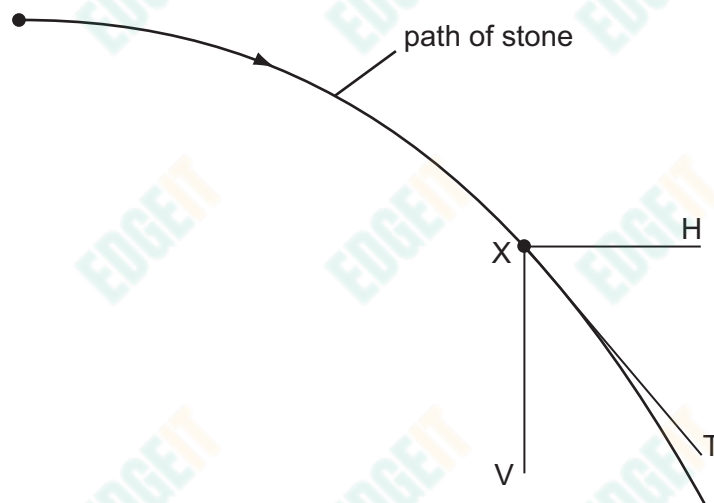
10 The gravitational field strength on the surface of planet P is one tenth of that on the surface of planet Q.

On the surface of P, a body has its mass measured to be 1.0 kg and its weight measured to be 1.0 N.

What results are obtained for measurements of the mass and weight of the same body on the surface of planet Q?

	mass on Q	weight on Q
A	1.0 kg	0.1 N
B	1.0 kg	10 N
C	10 kg	10 N
D	10 kg	100 N

11 A stone is projected horizontally in a vacuum and moves along a path as shown. X is a point on this path. XV and XH are vertical and horizontal lines respectively through X. XT is the tangent to the path at X.



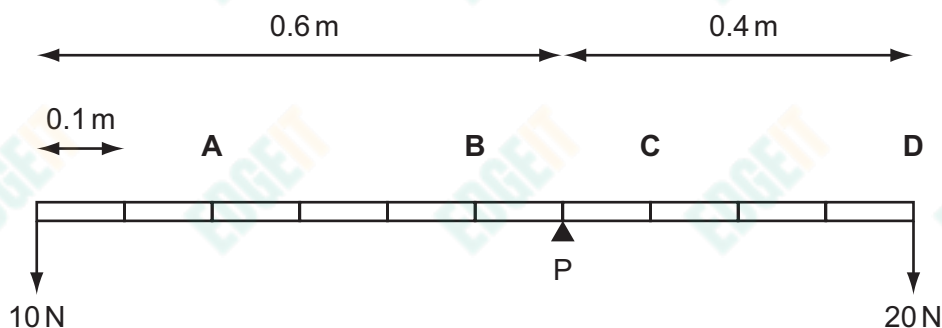
Along which direction or directions do forces act on the stone at X?

- A XV
- B XH
- C XV and XH
- D XT

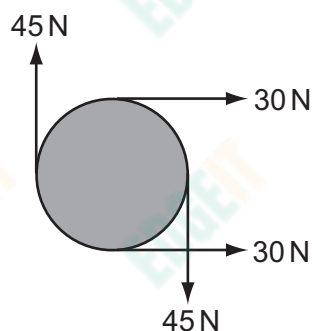
- 12 A uniform beam of weight 100 N is pivoted at P as shown. Weights of 10 N and 20 N are attached to its ends.

The length of the beam is marked off at 0.1 m intervals.

At which point should a further weight of 20 N be attached to achieve equilibrium?



- 13 The diagram shows four forces applied to a circular object.



Which of the following describes the resultant force and resultant torque on the object?

	resultant force	resultant torque
A	non-zero	non-zero
B	non-zero	zero
C	zero	non-zero
D	zero	zero

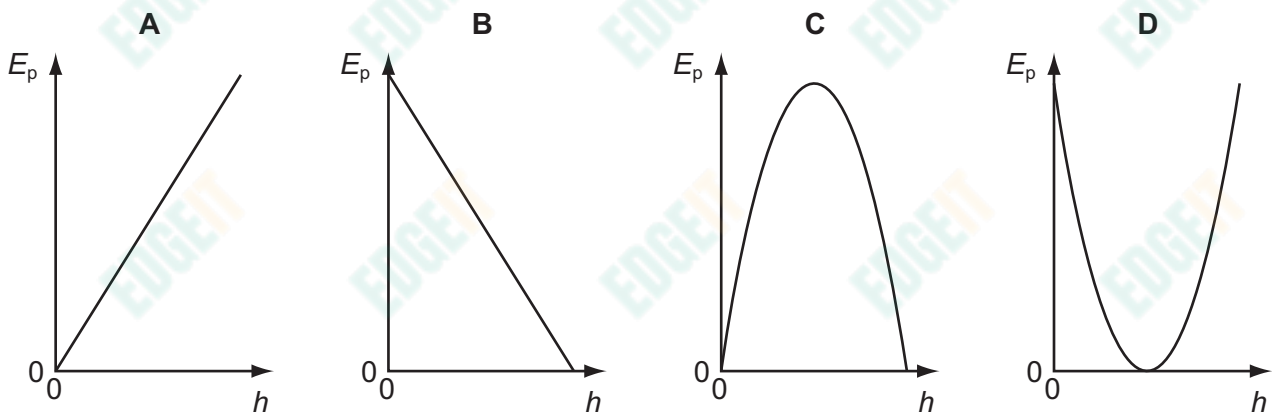
- 14 A car with a total mass of 1400 kg is travelling at 30 m s^{-1} .

What is the kinetic energy of the car?

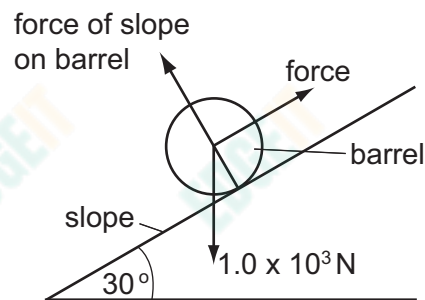
- A** 21 kJ **B** 42 kJ **C** 630 kJ **D** 1260 kJ

- 15 An object is thrown into the air.

Which graph shows how the potential energy E_p of the object varies with height h above the ground?



- 16 The diagram shows a barrel of weight $1.0 \times 10^3 \text{ N}$ on a frictionless slope inclined at 30° to the horizontal.



A force is applied to the barrel to move it up the slope at constant speed. The force is parallel to the slope.

What is the work done in moving the barrel a distance of 5.0 m up the slope?

- A** $1.0 \times 10^4 \text{ J}$ **B** $2.5 \times 10^3 \text{ J}$ **C** $4.3 \times 10^3 \text{ J}$ **D** $5.0 \times 10^3 \text{ J}$
- 17 Why does the pressure increase when a sealed container of gas is heated?
- A** The gas molecules collide more often with each other.
B The gas molecules expand when they are heated.
C The gas molecules travel faster and hit the walls of the container more often.
D There are more gas molecules present to collide with the walls of the container.

- 18 Liquids X and Y are stored in large open tanks. Liquids X and Y have densities of 800 kg m^{-3} and 1200 kg m^{-3} respectively.

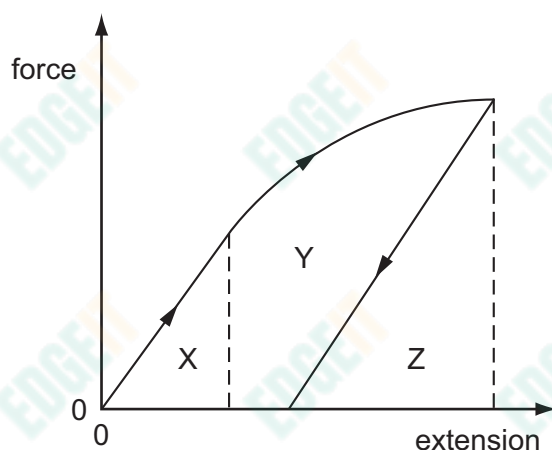
At what depths are the pressures equal?

	depth in liquid X	depth in liquid Y
A	8 m	12 m
B	10 m	10 m
C	15 m	10 m
D	18 m	8 m

- 19 When white sugar granules are heated, they melt. When the melt is cooled quickly, a brittle solid form of toffee is produced.

How does the structure of the sugar change?

- A** amorphous to polymeric
B crystalline to amorphous
C crystalline to polymeric
D polymeric to amorphous
- 20 A ductile material is stretched by a tensile force to a point beyond its elastic limit. The tensile force is then reduced to zero. The graph of force against extension is shown below.



Which area represents the net work done on the sample?

- A** X **B** X + Y **C** Y + Z **D** Z

- 21 A wire stretches 8 mm under a load of 60 N.

A second wire of the same material, with half the diameter and a quarter of the original length of the first wire, is stretched by the same load.

Assuming that Hooke's law is obeyed, what is the extension of this wire?

- A 1 mm B 4 mm C 8 mm D 16 mm

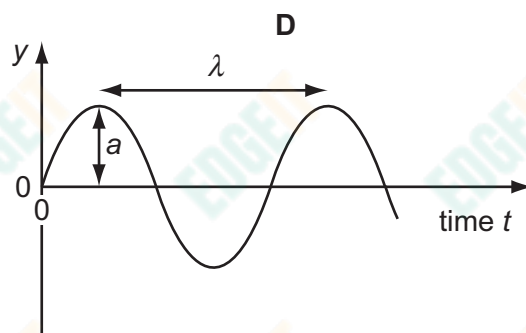
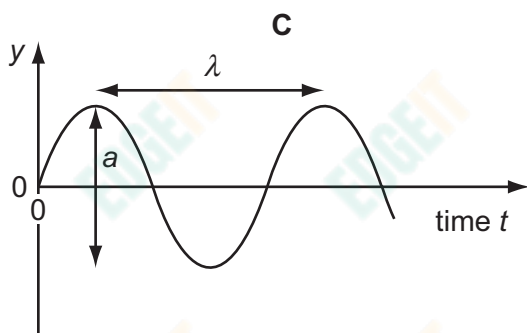
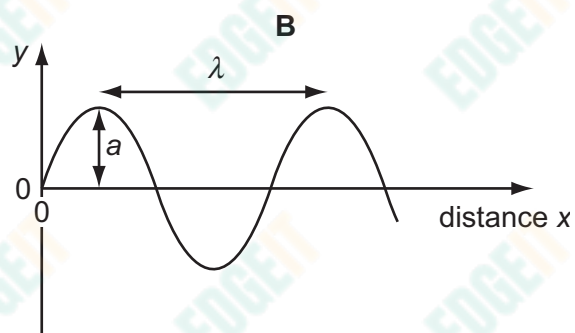
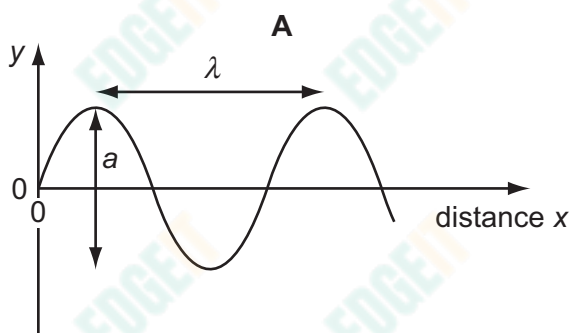
- 22 Polarisation is a phenomenon associated with a certain type of wave.

Which condition **must** be fulfilled if a wave is to be polarised?

- A It must be a light wave.
B It must be a longitudinal wave.
C It must be a radio wave.
D It must be a transverse wave.

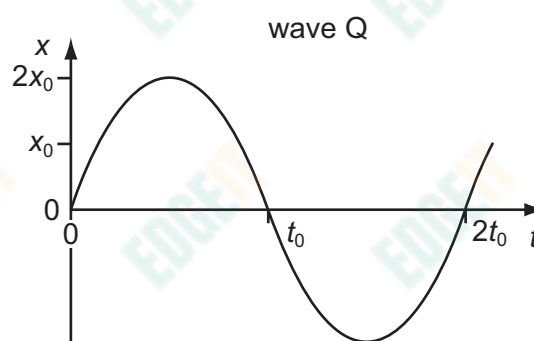
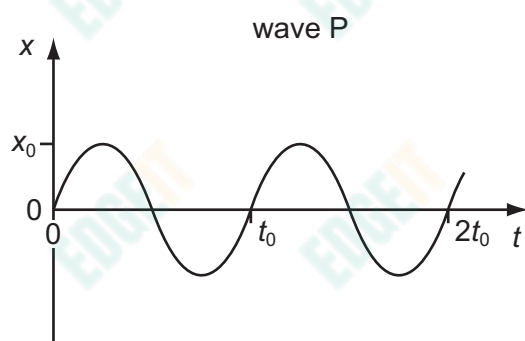
- 23 A sound wave has displacement y at distance x from its source at time t .

Which graph correctly shows the amplitude a and the wavelength λ of the wave?



- 24** The intensity of a progressive wave is proportional to the square of the amplitude of the wave. It is also proportional to the square of the frequency.

The variation with time t of displacement x of particles in a medium, when two progressive waves P and Q pass separately through the medium, are shown on the graphs.



The intensity of wave P is I_0 .

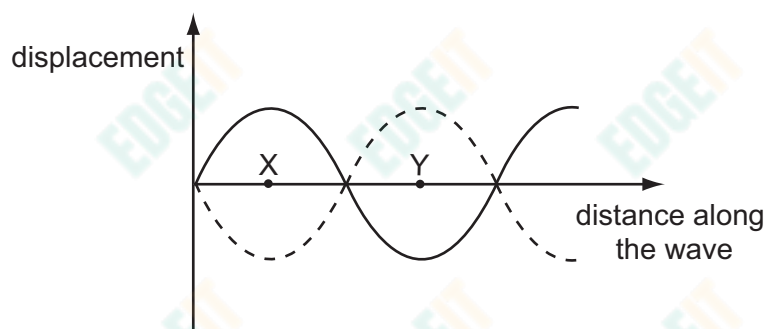
What is the intensity of wave Q?

- A** $\frac{1}{2}I_0$ **B** I_0 **C** $8I_0$ **D** $16I_0$
- 25** A sound wave of frequency 150 Hz travels in water at a speed of 1500 m s^{-1} . It then travels through the surface of the water and into air, where its speed is 300 m s^{-1} .

Which line in the table gives the correct values for the wavelengths of the sound in water and in air?

	wavelength in water/m	wavelength in air/m
A	0.10	0.10
B	0.10	0.50
C	10	2.0
D	10	50

- 26 The graph represents a standing wave at two different times.

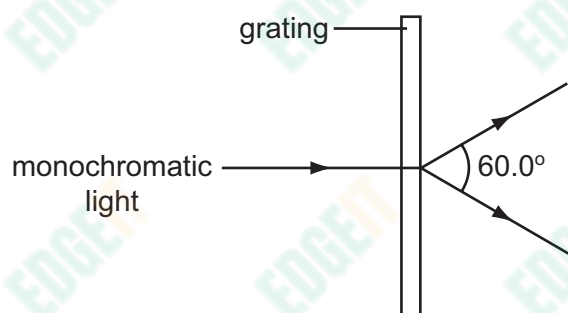


What does the distance XY represent?

- A half the amplitude
 - B half the frequency
 - C half the period
 - D half the wavelength
- 27 In which situation does diffraction occur?
- A A wave bounces back from a surface.
 - B A wave passes from one medium into another.
 - C A wave passes through an aperture.
 - D Waves from two identical sources are superposed.
- 28 Light of wavelength 700 nm is incident on a pair of slits, forming fringes 3.0 mm apart on a screen.
- What is the fringe spacing when light of wavelength 350 nm is used and the slit separation is doubled?
- A 0.75 mm
 - B 1.5 mm
 - C 3.0 mm
 - D 6.0 mm

- 29 A diffraction grating is used to measure the wavelength of monochromatic light.

The spacing of the slits in the grating is 1.15×10^{-6} m. The angle between the first order diffraction maxima is 60.0° , as shown in the diagram.

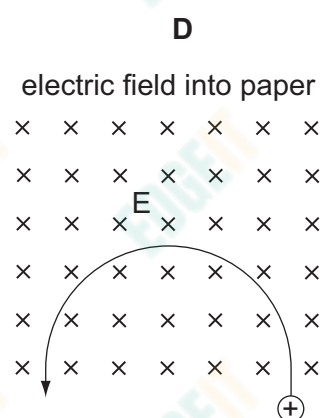
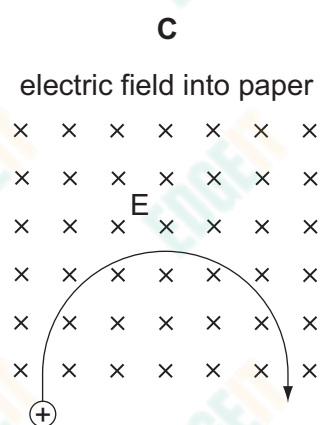
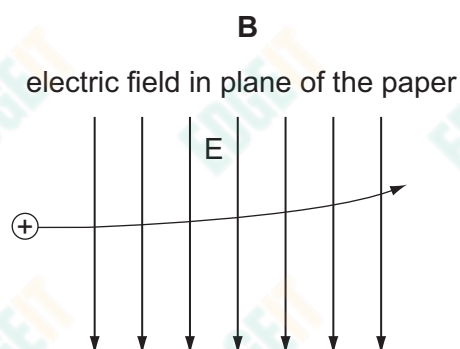
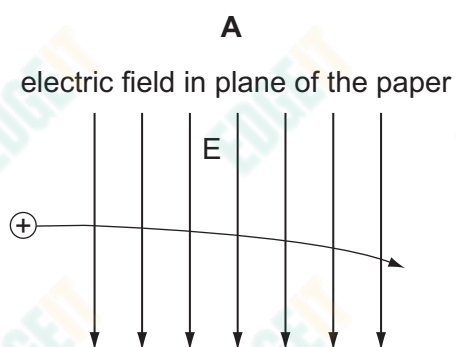


What is the wavelength of the light?

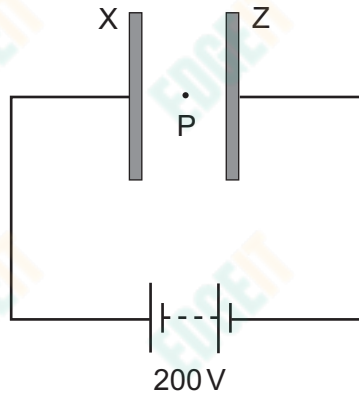
- A 287 nm B 498 nm C 575 nm D 996 nm

- 30 A positively charged particle is projected into a region of uniform electric field E .

Which diagram represents the motion of the particle in the electric field?



- 31 Two large parallel plates X and Z are placed 5.0 mm apart and connected as shown to the terminals of a 200 volt d.c. supply.

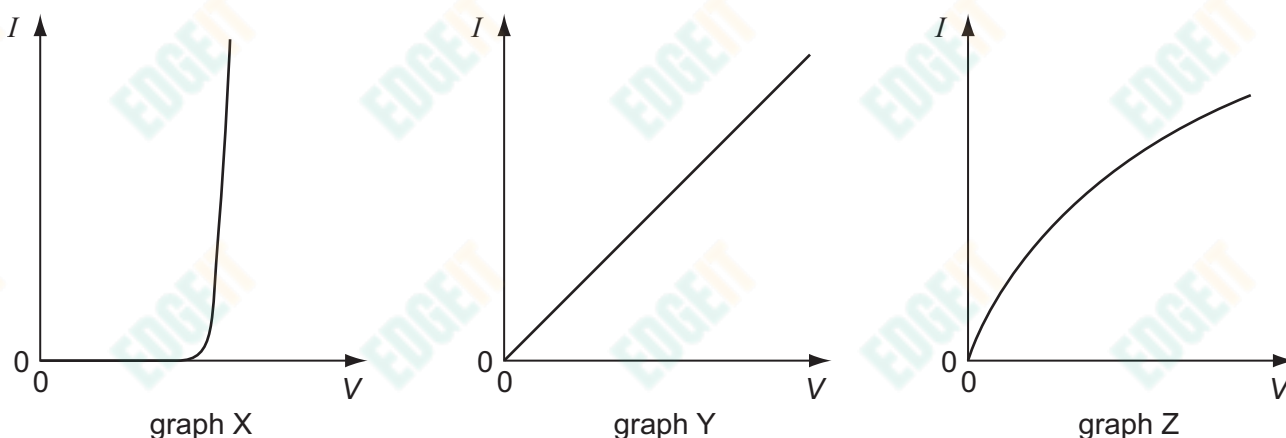


A small oil drop at P carries one excess electron.

What is the magnitude of the electrostatic force acting on the oil drop due to the electric field between the plates?

- A $6.4 \times 10^{-15} \text{ N}$
- B $6.4 \times 10^{-18} \text{ N}$
- C $1.6 \times 10^{-19} \text{ N}$
- D $4.0 \times 10^{-24} \text{ N}$

- 32 The graphs show the variation with potential difference V of the current I for three circuit components.

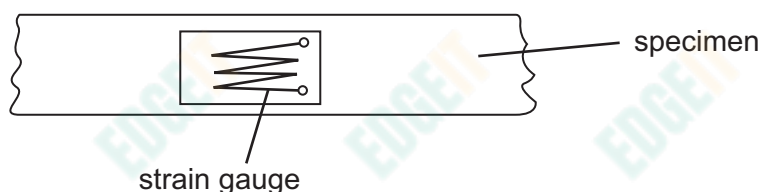


The components are a metal wire at constant temperature, a semiconductor diode and a filament lamp.

Which row of the table correctly identifies these graphs?

	metal wire at constant temperature	semiconductor diode	filament lamp
A	X	Z	Y
B	Y	X	Z
C	Y	Z	X
D	Z	X	Y

- 33 Tensile strain may be measured by the change in electrical resistance of a strain gauge. A strain gauge consists of folded fine metal wire mounted on a flexible insulating backing sheet. The strain gauge is firmly attached to the specimen, so that the strain in the metal wire is always identical to that in the specimen.

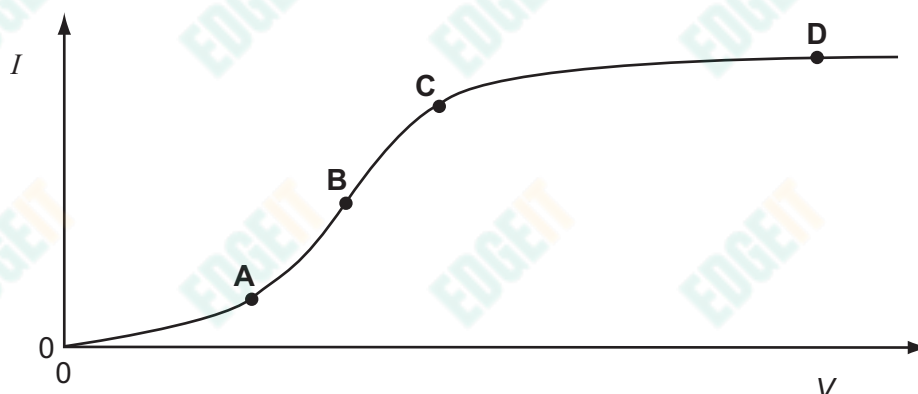


When the strain in the specimen is increased, what happens to the resistance of the wire?

- A** It decreases, because the length decreases and the cross-sectional area increases.
- B** It decreases, because the length increases and the cross-sectional area decreases.
- C** It increases, because the length decreases and the cross-sectional area increases.
- D** It increases, because the length increases and the cross-sectional area decreases.

- 34 The graph shows how the electric current I through a conducting liquid varies with the potential difference V across it.

At which point on the graph does the liquid have the smallest resistance?

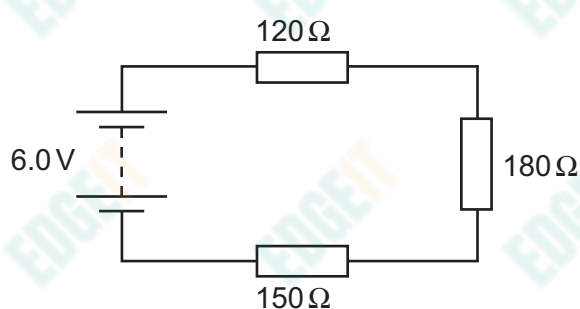


- 35 An electrical component has the following circuit symbol.



What does this symbol represent?

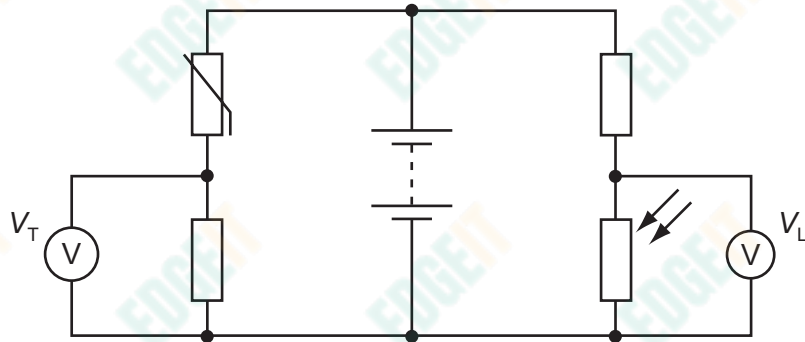
- A variable resistor (rheostat)
 B fuse
 C light-dependent resistor
 D thermistor
- 36 Three resistors are connected in series with a battery as shown in the diagram. The battery has negligible internal resistance.



What is the potential difference across the $180\ \Omega$ resistor?

- A 1.6 V B 2.4 V C 3.6 V D 6.0 V

- 37 In the circuit below, the reading V_T on the voltmeter changes from high to low as the temperature of the thermistor changes. The reading V_L on the voltmeter changes from high to low as the level of light on the light-dependent resistor (LDR) changes.



The readings on V_T and V_L are both high.

What are the conditions of temperature and light level?

	temperature	light level
A	low	low
B	low	high
C	high	low
D	high	high

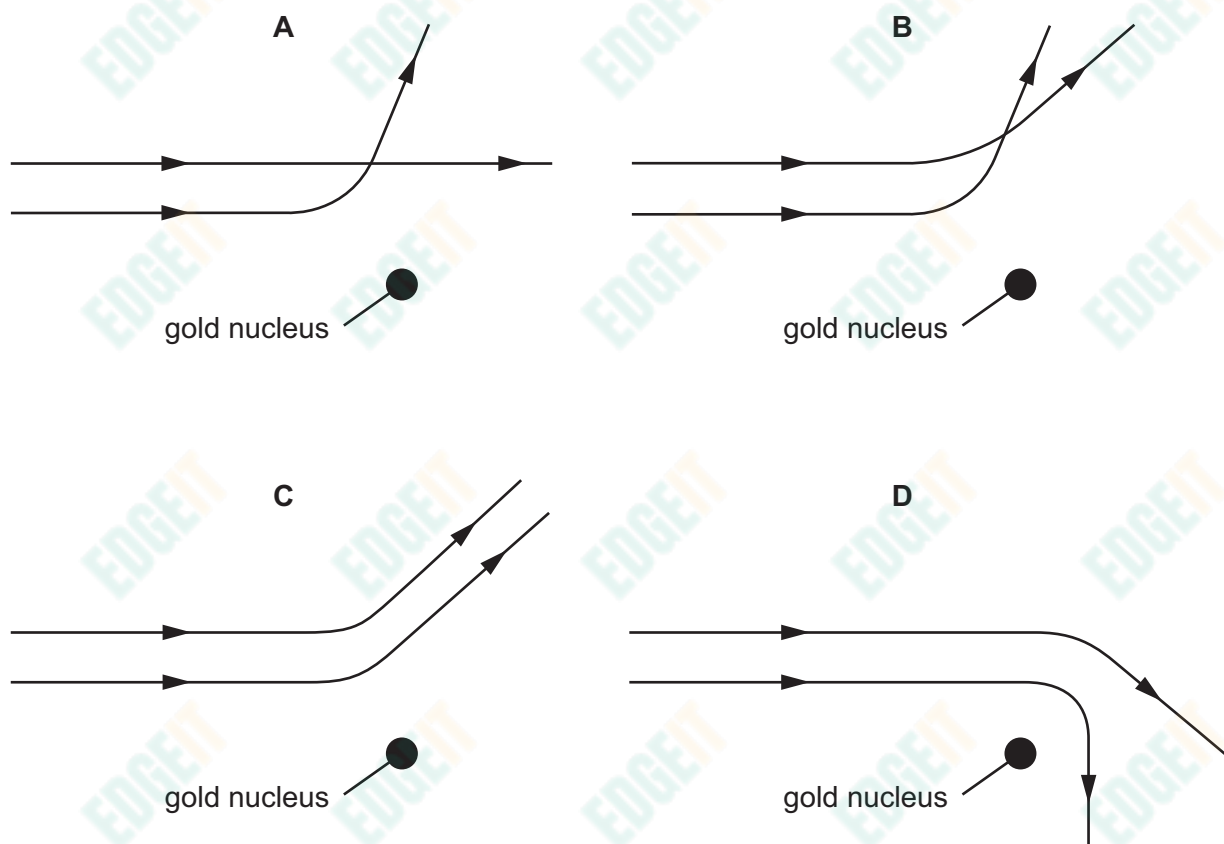
- 38 An atomic nucleus emits a β -particle.

What change does this cause to the proton and nucleon numbers of the nucleus?

	proton number	nucleon number
A	-1	+1
B	0	-1
C	+1	-1
D	+1	0

39 Two α -particles with equal energies are fired towards the nucleus of a gold atom.

Which diagram best represents their paths?



40 A nuclear reaction is represented by the equation



What is particle X?

- A an α -particle
- B a β -particle
- C a neutron
- D a proton

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